CAPSTONE PROJECT REPORT

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Course: Deep Learning Architectures and Techniques

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1. Introduction

Objective of the Project

The objective of this project is to implement and analyze different deep learning models including Feedforward Neural Networks (FNN), Convolutional Neural Networks (CNN), and Long Short-Term Memory Networks (LSTM) on different types of datasets such as tabular, image, and time-series data.

Brief Explanation of Each Experiment

- Experiment 1: FNN for income classification (tabular data)
- Experiment 2: CNN for image classification (clothing dataset)
- Experiment 3: LSTM for temperature forecasting (time-series data)

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2. Methodology

Experiment 1: FNN

Dataset Description

Dataset used: UCI Adult Dataset

Dataset Link: https://www.kaggle.com/code/jyoti06/uci-adult-dataset/input

Preprocessing Steps

- Handled missing values
- Encoded categorical variables
- Normalized features
- Train-test split

Model Architecture

Input → Dense (64, ReLU) → Dense (32, ReLU) → Output (Sigmoid)

Experiment 2: CNN

Dataset Description

Dataset used: Fashion-MNIST

Dataset Link: Built-in dataset available in TensorFlow/Keras

Preprocessing Steps

- Normalized pixel values
- Reshaped images
- Train-validation split

Model Architecture

Conv → Pool → Conv → Pool → Dense → Output

Experiment 3: LSTM

Dataset Description

Dataset used: Delhi Climate dataset from Kaggle

Dataset Link: https://www.kaggle.com/datasets/sumanthvrao/daily-climate-time-series-data

Preprocessing Steps

- Removed missing values
- Sorted data by date
- Normalized data
- Created sequences

Model Architecture

LSTM → LSTM → Dense

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3. Implementation Details

Framework Used

- TensorFlow / Keras
- Python

Hyperparameters

Model	Epochs	Batch Size	Optimizer	Loss
FNN	20	32	Adam	Binary Crossentropy
CNN	10	64	Adam	Sparse Categorical Crossentropy
LSTM	10	32	Adam	Mean Squared Error

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4. Results

Experiment 1: FNN

Accuracy: Test Accuracy: 0.8514096140861511

The model achieved good performance on tabular data classification.

Figure 1: Training vs Validation Loss Curve

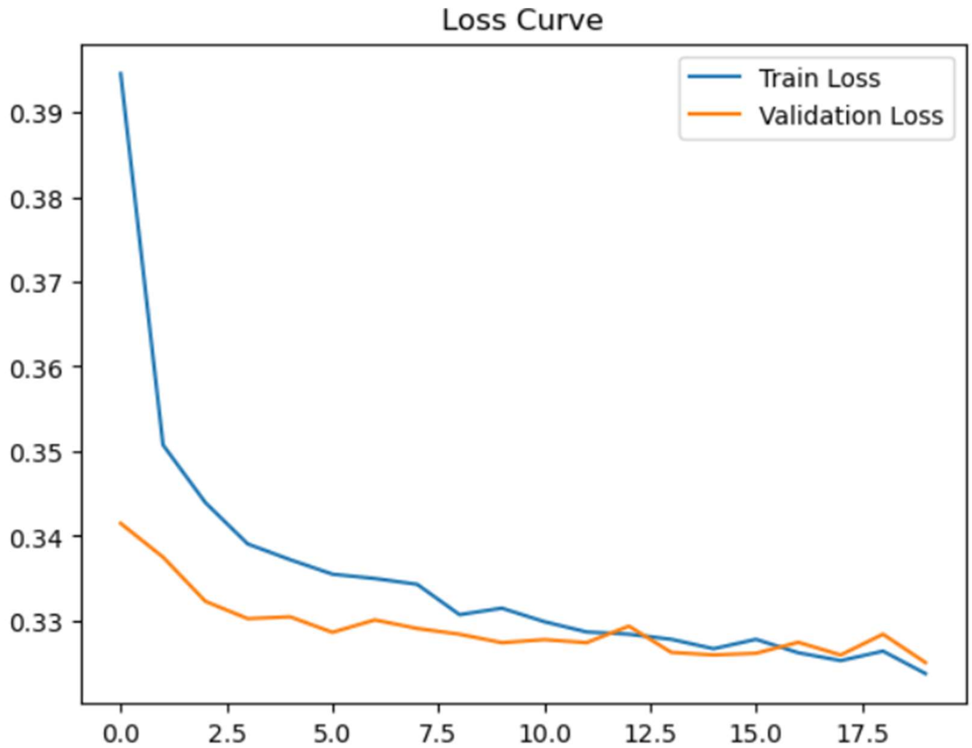
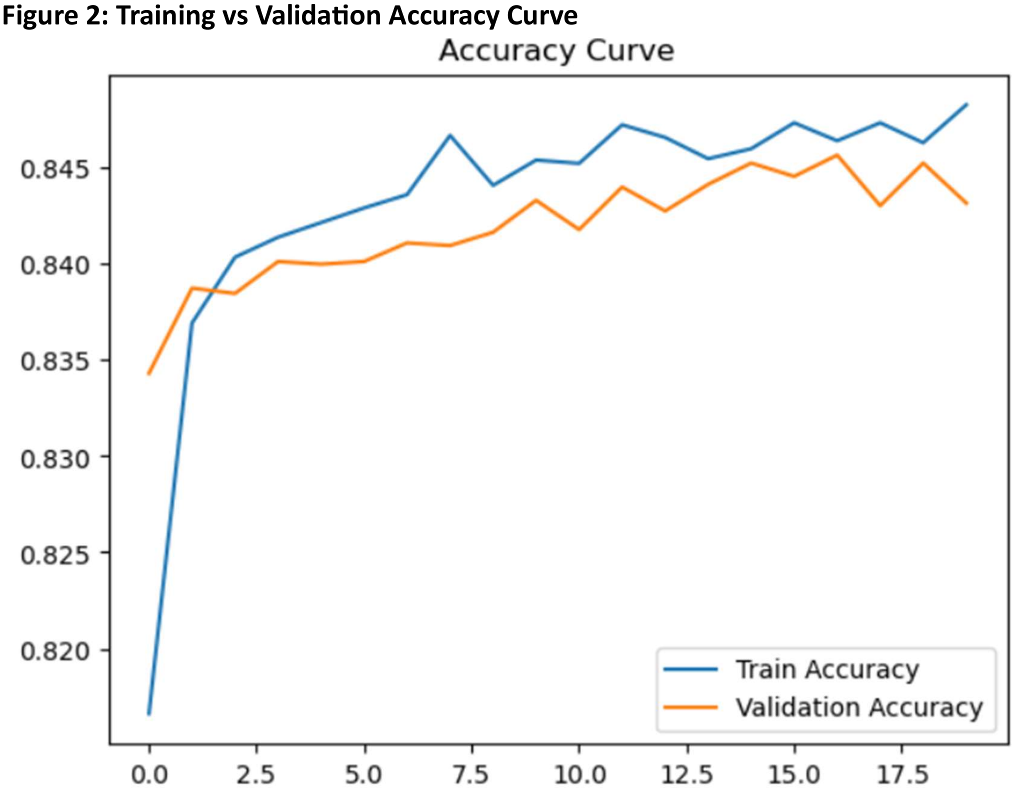


Figure 2: Training vs Validation Accuracy Curve



Experiment 2: CNN

Accuracy: Test Accuracy: 0.9075000286102295

The CNN model performed well in image classification tasks.

Figure 3: Training vs Validation Accuracy Curve

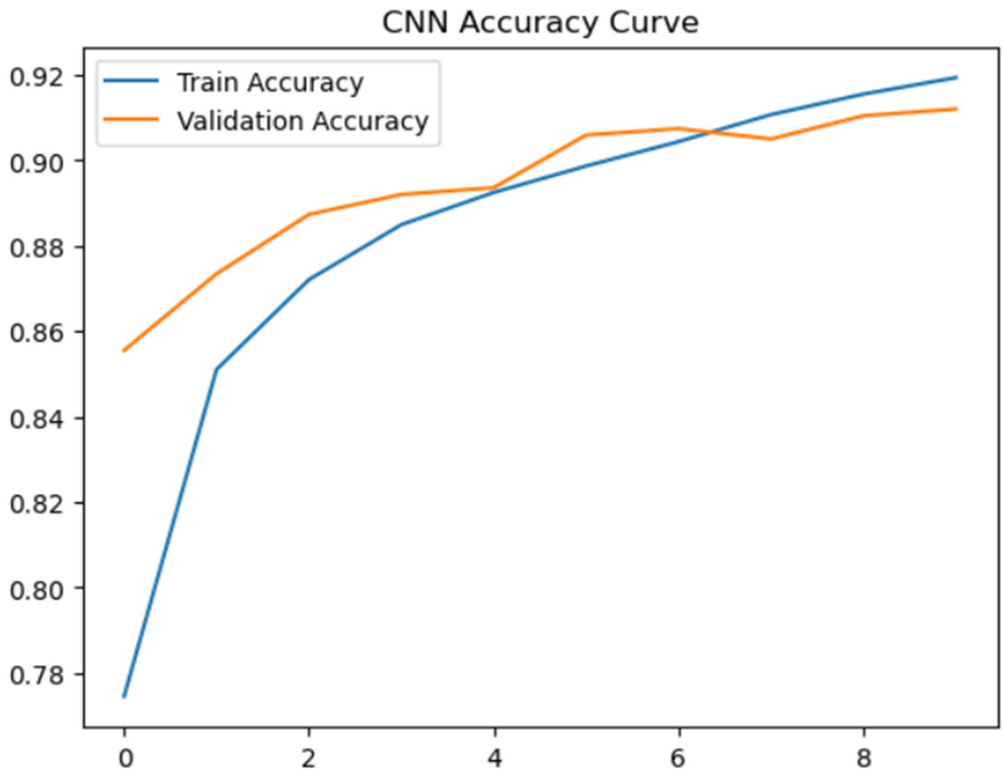


Figure 4: Training vs Validation Loss Curve

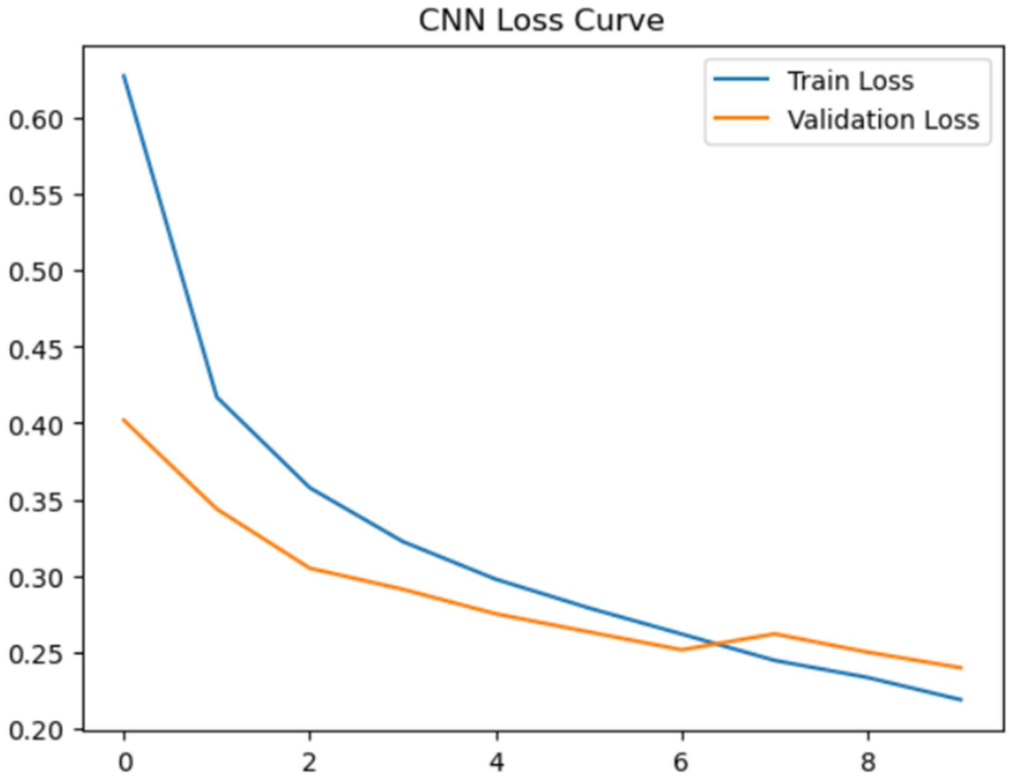


Figure 5: Evaluation Metrics

	precision	recall	f1-score	support
0	0.83	0.88	0.86	1000
1	1.00	0.97	0.98	1000
2	0.83	0.89	0.86	1000
3	0.91	0.91	0.91	1000
4	0.83	0.89	0.86	1000
5	0.98	0.98	0.98	1000
6	0.78	0.64	0.70	1000
7	0.96	0.96	0.96	1000
8	0.98	0.98	0.98	1000
9	0.97	0.97	0.97	1000
accuracy			0.91	10000
macro avg	0.91	0.91	0.91	10000
weighted avg	0.91	0.91	0.91	10000

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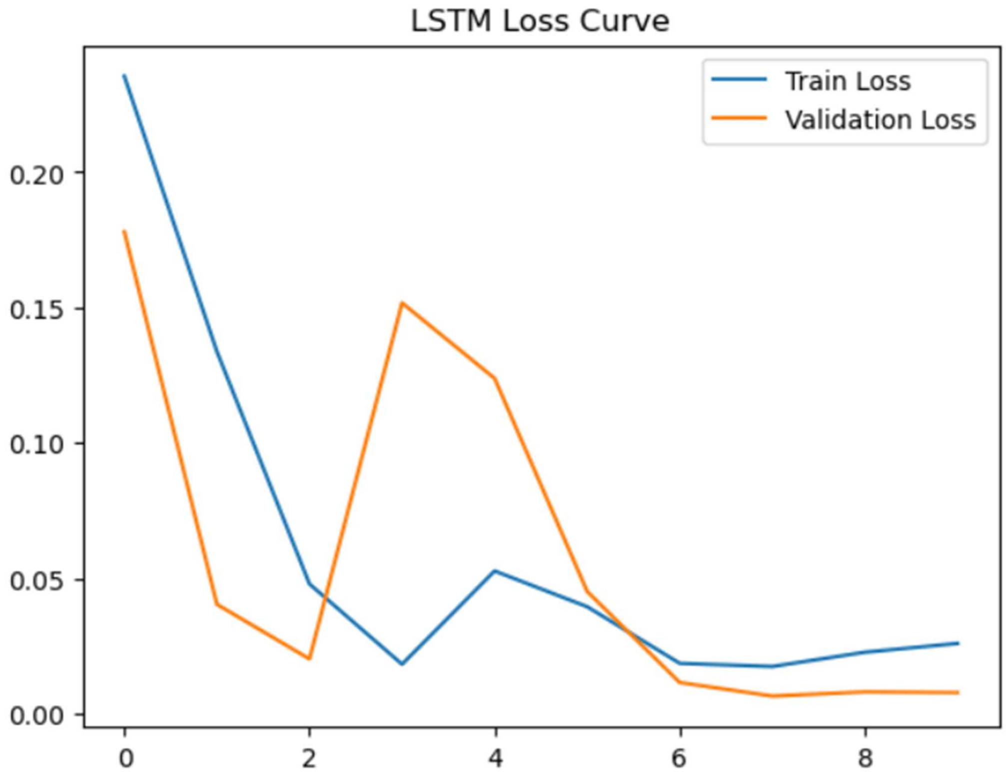
[[880   0  20  17   3   1  72   0   7   0]
 [  2 970   0  21   4   0   1   0   2   0]
 [ 14   0 886  11  50   0  38   0   1   0]
 [ 15   2  11 911  27   1  30   0   3   0]
 [  1   1  54  20 891   0  32   0   1   0]
 [  0   0   0   0   0 984   0  13   0   3]
 [142   0  88  25  94   0 640   0  11   0]
 [  0   0   0   0   0  12   0 961   0  27]
 [  1   1   7   1   2   1   3   3 981   0]
 [  0   0   0   0   0   7   1  21   0 971]]
```

Experiment 3: LSTM

Test MSE: 0.04150710999965668

The low MSE indicates accurate prediction of temperature trends.

Figure 6: Training vs Validation Loss Curve



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5. Discussion

Observations

- FNN performed well on structured tabular data
- CNN achieved high accuracy for image classification
- LSTM captured temporal dependencies effectively

Challenges Faced

- Data preprocessing complexity
- Hyperparameter tuning
- Avoiding overfitting

Comparison of Models

Model	Best Use Case
FNN	Tabular data
CNN	Image data
LSTM	Time-series data

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6. Conclusion

This project successfully demonstrated the implementation of different deep learning models across multiple domains. Each model performed effectively for its respective task.

Key Learnings

- Importance of preprocessing
- Model selection based on data type
- Understanding different deep learning architectures

Real-World Applications

- Income prediction systems
- Image recognition systems
- Weather forecasting

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7. References

- UCI Adult Dataset
- Fashion-MNIST
- Kaggle
- TensorFlow Documentation